

SAHIEL BOSE

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EDUCATION

University of California, Los Angeles

Los Angeles, CA

B.S. in Mathematics of Computation

EXPERIENCE

Optivia

Dublin, CA

Co-Founder

March 2025 – Present

- Lead the design and development of AI agents for the Optivia prototype, serving as the primary architect of its core intelligence, agentic workflows, and decision-making logic.
- Designed the mathematical modeling and algorithmic logic behind the multi-agent pipeline, including prompt routing, complexity scoring, model selection, and prompt engineering (patent pending).
- Built a framework that intercepts user inputs, dynamically enriches prompts with targeted clarifying questions, and activates Claude Code commands using multi-agent systems and the Anthropic SDK.

ThinkNeuro LLC

Dublin, CA

Neurotechnology & Software Engineering Intern

November 2024 – Present

- Developing a mobile application using React Native and JavaScript for neuroscience research applications.
- Creating a machine learning model to reduce false detections in opioid overdose prediction.
- Conducting research and co-authoring a paper on opioid overdose and its research applications.

UC Santa Cruz AIEA Lab

Santa Cruz, CA

AI Research Intern

October 2024 – April 2026

- Research logical reasoning in large language models (LLMs) and study methods to enhance their problem-solving capabilities.
- Develop and implement structured exercises to improve LLMs' ability to perform logical inference and reasoning tasks.
- Explore algorithmic techniques to refine and optimize LLM-generated reasoning processes.

UC Riverside Research Lab

Riverside, CA

Machine Learning Research Intern

May 2025 – January 2026

- Assisted a Ph.D. student on a Graph Neural Network (GNN)-based project aimed at detecting traffic congestion hotspots in large-scale road networks using AI and spatial data analytics.
- Supported data preprocessing, model training, and optimization to improve insights for urban planning.
- Collaborated on research involving real-world infrastructure and transportation systems.

PROJECTS

Opioid Technology Device + Application | *ML, Neurotechnology*

November 2024 – Present

- Contribute to the development of a neurotechnology device for opioid overdose prediction using machine learning.
- Support hardware-software integration and research paper development.
- Worked on a project with a patent application and code copyright in progress.

UC Santa Cruz ProSLM Project | *LLM Systems, Deployment*

June 2025 – January 2026

- Assist with website hosting and integration for the ProSLM project.
- Help connect different LLM research components into a unified platform and support deployment and user access.

UC Riverside HotSpot Detection with GNNs | *Graph Neural Networks*

May 2025 – January 2026

- Assisted in developing GNN models to identify traffic congestion hotspots using real-time geospatial data.
- Tested smaller benchmark datasets to validate model performance for transportation analysis.

ACSEF Project: Sign Language Recognition System | *CNNs, LSTMs*

December 2024 – March 2025

- Co-led the design of an AI-powered ASL recognition system to help bridge communication gaps for Deaf and hard-of-hearing users.
- Built the system using CNNs, LSTMs, and Raspberry Pi-based deployment.
- Earned 2nd place at the Alameda County Science & Engineering Fair.

TECHNICAL SKILLS

AI Techniques: Logical Reasoning, Linear Regression, KNN, Deep Learning, Neural Networks, Agentic Workflows, Prompt Engineering, Multi-Agent Systems, Model Training

Frameworks & Libraries: TensorFlow, PyTorch, Keras, React Native, n8n, Anthropic SDK

AI Applications & Models: CNNs, RNNs, GNNs, GANs, LLMs, NLP, Computer Vision, AI Agents, Prompt Routing

Languages: Python